

Unit I

16/4

Curve fitting →

- Method of finding suitable relation in form of $y = f(x)$ for set of observed values $(x_i, y_i) \forall i = 1, 2, \dots, n$

Method of least squares →

- Let $y = f(x)$ be approx. relation fitting to given data $(x_i, y_i) \forall i = 1, 2, \dots, n$
- Difference $R_i = Y_i - y_i \rightarrow$ Residual / Estimated errors
- $Y_i = f(x_i) \rightarrow$ Estimated / Expected value
- $y_i \rightarrow$ Observed value
- Provides relationship $y = f(x)$ such that sum of squares of residual is minimum

Fitting straight line of form $y = ax + b$ →Consider n data points $(x_i, y_i) \forall i = 1, 2, \dots, n$

$$R_i = (ax_i + b) - y_i$$

$$R = (ax + b) - y$$

$$\text{Let } S = \sum R^2 = \sum [(ax + b) - y]^2$$

Necessary conditions for S to be minimum:

$$* \frac{\partial S}{\partial a} = 0 \Rightarrow \sum 2 [(ax + b) - y] x = 0 \Rightarrow \sum ax^2 + \sum bx - \sum xy = 0$$

$$\sum xy = a \sum x^2 + \sum bx \Rightarrow \sum y = a \sum x + bn$$

$$* \frac{\partial S}{\partial b} = 0 \Rightarrow \sum 2 [(ax + b) - y] \times 1 = 0 \Rightarrow \sum ax + \sum b - \sum y = 0$$

$$\sum y = a \sum x + bn$$

$$\begin{aligned} \sum y &= a \sum x + bn \\ \sum xy &= a \sum x^2 + \sum bx \end{aligned} \quad] \rightarrow (i) \rightarrow \text{Normal eqns.}$$

By solving (i), we get a & b

Normal eqns. for $y = ax + b \rightarrow$

$$\begin{aligned} \sum y &= a \sum x + bn \\ \sum xy &= a \sum x^2 + b \sum x \end{aligned}$$

Normal eqns. for $y = ax^2 + bx + c \rightarrow$

$$\begin{aligned} \sum y &= a \sum x^2 + b \sum x + cn \\ \sum xy &= a \sum x^3 + b \sum x^2 + c \sum x \\ \sum x^2 y &= a \sum x^4 + b \sum x^3 + c \sum x^2 \end{aligned}$$

Normal eqns. for $y = ax^2 + b \rightarrow$

$$\begin{aligned} \sum y &= a \sum x^2 + bn \\ \sum xy &= a \sum x^3 + b \sum x \end{aligned}$$

Note Depends on no. of unknowns, not degree

Normal eqns. for $y = ae^{bx} \rightarrow$

Take log on both sides $\rightarrow \log y = \log a + \log e^{bx} \Rightarrow Y = A + bX$

$$\begin{aligned} \text{NE for (i)} &\rightarrow \sum Y = An + b \sum x \\ &\quad \sum xY = A \sum x + b \sum x^2 \end{aligned} \quad] \rightarrow (ii)$$

By solving (ii), we get A & b
To get $a \Rightarrow a = e^A$

NE for $y = ax^b$ (geometric curve) $\rightarrow$

$$\log y = \log a + b \log x \Rightarrow Y = A + bX$$

$$\sum Y = An + b \sum X$$

$$\sum XY = A \sum X + b \sum X^2 \quad \sum XY = A \sum X + b \sum X^2$$

$$a = e^A, \Rightarrow \text{to find } a$$

NE for $y = ax^n + b$ [n is constant] $\rightarrow$

$$\text{Take } X = x^n \Rightarrow y = aX + b$$

$$\sum y = a \sum X + bn$$

$$\sum xy = a \sum X^2 + b \sum X$$

18/4 NE for $y = a + \frac{b}{x}$ $\rightarrow$

$$\text{Take } X = 1/x \Rightarrow y = a + bX$$

$$\sum y = an + b \sum X$$

$$\sum xy = a \sum X + b \sum X^2$$

NE for $y = ax + b/x$ $\rightarrow$

$$R = (ax + b/x) - y \Rightarrow S = \sum R^2 = \sum (ax + b/x - y)^2$$

$$\frac{\partial S}{\partial a} = 0 = 2 \left((ax + b/x) - y \right) \cdot x \Rightarrow \sum xy = a \sum x^2 + bn$$

$$\frac{\partial S}{\partial b} = 0 = 2 \left((ax + b/x) - y \right) \left(1/x \right) = 2 \sum (a + b/x^2 - y/x)$$

$$\sum \frac{y}{x} = an + b \sum \frac{1}{x^2}$$

Q Fit straight line of form $y = a + bx$ for:

x	1	2	3	4
y	3	5	8	10

Ans. NE $\rightarrow \sum y = an + b \sum x$

$$\sum xy = a \sum x + b \sum x^2$$

$$\sum x = 10, \sum y = 26, \sum xy = 77, \sum x^2 = 30$$

$$26 = 4a + 10b$$

$$\Rightarrow a = 0.5, b = 2.4$$

$$77 = 10a + 30b$$

Required curve $\rightarrow y = a + bx = 0.5 + 2.4x$

Q Fit parabola of form $y = ax^2 + bx + c$ for:

x	1	2	3	4
y	3	5	8	10

Ans. NE $\rightarrow \sum y = a \sum x^2 + b \sum x + cn$

$$\sum xy = a \sum x^3 + b \sum x^2 + c \sum x$$

$$\sum x^2 y = a \sum x^4 + b \sum x^3 + c \sum x^2$$

$$\sum x = 10, \sum y = 26, \sum xy = 77, \sum x^2 = 30, \sum x^3 = 100, \sum x^4 = 354$$

$$\sum x^2 y = 255$$

$$26 = 30a + 10b + 4c$$

$$77 = 100a + 30b + 10c$$

$$\Rightarrow a = 0, b = 2.4, c = 0.5$$

$$255 = 354a + 100b + 30c$$

Required curve $\rightarrow y = 2.4x + 0.5$

Q Fit exponential curve of form $y = ae^{bx}$ for:

x	1	2	3	4
y	3	5	8	10

Ans. $\log y = \log a + \log e^{bx} \Rightarrow Y = A + bx$

NE $\rightarrow \sum Y = An + b \sum x$
 $\sum xY = A \sum x + b \sum x^2$

x	1	2	3	4
y	3	5	8	10
Y	1.098	1.609	2.079	2.302 [ln]

$\sum x = 10, \sum y = 26, \sum Y = 7.09, \sum xY = 19.76, \sum x^2 = 30$

$7.09 = 4A + 10b$

$\Rightarrow A = 0.75 \Rightarrow a = e^A = 2.12, b = 0.407$

$19.76 = 10A + 30b$

Required curve $\rightarrow y = 2.12e^{0.407x}$

Q On avg., surface area A of human beings is related to weight W & height H. Measurements on individuals of height 180cm and diff. Ws gives values of A as:

W(Kg)	70	75	77	80	82
A(m ²)	2.10	2.12	2.15	2.20	2.22

(i) Fit curve of form $A = aW^b$

(ii) Predict A for W = 95 Kg

Ans. $A = aW^b$ [A, W $\rightarrow$ variables]

$\log A = \log a + b \log W \Rightarrow Y = m + bX$

NE $\Rightarrow \sum Y = mn + b \sum X$

$\sum XY = m \sum X + b \sum X^2$

W	70	75	77	80	82
$X = \log W$	4.248	4.317	4.343	4.382	4.406
A	2.10	2.12	2.15	2.20	2.22
$Y = \log A$	0.741	0.751	0.765	0.788	0.797

$$\sum X = 21.698, \sum Y = 3.844, \sum X^2 = 94.158, \sum XY = 16.676$$

$$3.844 = 5m + 21.698b \quad \Rightarrow m = \frac{4.339}{-8.141}, b = \frac{9.4 \times 10^{-5}}{-2.053}$$

$$16.676 = 21.698m + 94.158b$$

Q Find law of form $y = ax + b/x$ for:

x	2	3	4	5
y	5	8	10	11

Ans. $\sum xy = a \sum x^2 + b \sum 1/x$

$$\sum y/x = a \sum n + b \sum 1/x^2$$

x	2	3	4	5
y	5	8	10	11
x^2	4	9	16	25
$1/x^2$	$1/4$	$1/9$	$1/16$	$1/25$
y/x	$5/4$	$8/9$	$5/8$	$11/25$

$$129 = 54a + 4b$$

$$9.867 = 4a + 0.463b$$

$$9.867$$

$$\Rightarrow a = \frac{2.250}{-5.211}, b = \frac{1.861}{-38.1}$$

19/4 Correlation & Regression →

Let $\{x_i (i = 1, 2 \dots n)\}$ be given data

Mean $\mu = \bar{x} = \frac{\sum x_i}{n}$

Median Middle term of given data

Mode Most repeated term

Std. Dev $\sigma_x = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$ $\sigma_x = \sqrt{\frac{\sum x^2 - n\bar{x}^2}{n}}$

Variance σ^2

Correlation → Changes in one variable are changes in another variable

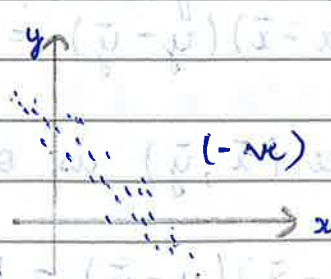
↳ Positive → $\uparrow$ (or $\downarrow$) in one variable $\Rightarrow \uparrow$ ($\downarrow$) in another variable

↳ Negative → $\uparrow$ (or $\downarrow$) in one variable $\Rightarrow \downarrow$ (or $\uparrow$) in another variable

Let (x, y) be given data $\Rightarrow X = x - \bar{x}$ & $Y = y - \bar{y}$

$\sum XY > 0 \Rightarrow +ve$ correlation

$\sum XY < 0 \Rightarrow -ve$ correlation



$\frac{\sum XY}{n \sigma_x \sigma_y} = r \Rightarrow$ Coefficient of correlation

Methods to calculate $r \rightarrow$

$$\begin{aligned} \bullet \quad r &= \frac{\sum XY}{n \sigma_x \sigma_y} & \bullet \quad r &= \frac{\sum XY}{\sqrt{\sum X^2 \sum Y^2}} & \bullet \quad r &= \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{xy}^2}{2 \sigma_x \sigma_y} \\ \bullet \quad r &= \frac{n \sum XY - \sum X \sum Y}{(n \sum X^2 - (\sum X)^2)(n \sum Y^2 - (\sum Y)^2)} \end{aligned}$$

LoR Lines of regression $\rightarrow$

- Lines of best fit for given distribution suggest linear relationship between variables x & y
- Line gives best possible mean value of y for each specified value of $x \Rightarrow$ LoR of y on $x \Rightarrow y = a + bx$ (i)

Similarly, LoR of x on $y \Rightarrow x = A + By$

$$\begin{aligned} \text{NE of } y = a + bx &\rightarrow \sum y = an + b \sum x \quad \dots (ii) \\ \sum xy &= a \sum x + b \sum x^2 \quad \dots (iii) \end{aligned}$$

$$\text{Divide by } n \rightarrow \frac{\sum y}{n} = a + \frac{b}{n} \sum x \Rightarrow \bar{y} = a + b\bar{x}$$

$\therefore$ LoR always passes through point $(\bar{x}, \bar{y})$

Shift origin to $(\bar{x}, \bar{y}) \rightarrow$

$$\bullet (iii) \Rightarrow \sum (x - \bar{x})(y - \bar{y}) = a \sum (x - \bar{x}) + b \sum (x - \bar{x})^2$$

Since $(\bar{x}, \bar{y})$ is origin in XY plane, constant term $a = 0$

$$\sum (x - \bar{x})(y - \bar{y}) = b \sum (x - \bar{x})^2$$

$$\therefore b = \frac{\sum XY}{\sum X^2}$$

Since $\sum XY = n a_x a_y$ & $\sum X^2 = n a_x^2$

$$\therefore b_{yx} = \frac{n a_x a_y}{n a_x^2} = \frac{a_y}{a_x}$$

LoR of y on x will take form:

$$y - \bar{y} = \frac{a_y}{a_x} (x - \bar{x})$$

Similarly, $b_{xy} = \frac{\sum XY}{\sum Y^2} = \frac{n a_x}{n a_y^2} \Rightarrow x - \bar{x} = \frac{a_x}{a_y} (y - \bar{y}) \Rightarrow$ LoR of x on y

b Regression coefficient = $\theta = \dots$

Geometric Mean of regression coefficient of y on x & x on $y \rightarrow$

$$\sqrt{b_{xy} \times b_{yx}} = \sqrt{\frac{a_x}{a_y} \times \frac{a_y}{a_x}} = \sqrt{1} = 1 = \text{Coeff of correlation}$$

Wkt $r = \frac{\sum XY}{\sqrt{\sum X^2 \sum Y^2}}$

By Schwartz's inequality $\rightarrow (\sum a_i b_i)^2 \leq \sum a_i^2 \times \sum b_i^2$

$$(\sum XY)^2 \leq \sum X^2 \times \sum Y^2 \Rightarrow \frac{(\sum XY)^2}{\sum X^2 \sum Y^2} \leq 1 \Rightarrow r^2 \leq 1 \Rightarrow -1 \leq r \leq 1$$

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Q If θ is angle b/w LoR, then $\tan \theta = \frac{1-r^2}{r} \frac{a_x a_y}{a_x^2 + a_y^2}$

Ans. Consider LoR of y on $x \rightarrow y - \bar{y} = \frac{a_y}{a_x} (x - \bar{x})$

$$m_1 = \frac{a_y}{a_x}$$

Consider LoR of x on $y \rightarrow x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$

$$m_2 = \frac{1}{r} \frac{\sigma_y}{\sigma_x}$$

$$\text{Angle b/w LoR} \rightarrow \tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{r \frac{\sigma_y}{\sigma_x} - \frac{1}{r} \frac{\sigma_y}{\sigma_x}}{1 + r \frac{\sigma_y}{\sigma_x} \cdot \frac{1}{r} \frac{\sigma_y}{\sigma_x}}$$

$$\frac{\sigma_y}{\sigma_x} \left[\frac{r - 1/r}{1 + \sigma_y^2 / \sigma_x^2} \right] = \left(\frac{\sigma_y}{\sigma_x} \frac{\sigma_x^2}{\sigma_x^2 + \sigma_y^2} \right) \left[\frac{r^2 - 1}{r} \right]$$

$$\therefore \tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1 - r^2}{r} \right)$$

Case I If $r = 0 \Rightarrow \tan \theta \rightarrow \infty \Rightarrow \theta = \pi/2$
LoR are $\perp \Rightarrow$ No correlation b/w x & y

Case II If $r = \pm 1 \Rightarrow \tan \theta = 0 \Rightarrow \theta = 0$
LoR coincide / parallel $\Rightarrow$ perfect correlation b/w x & y

Q Regression eqns. of x & y are $y = 0.3x + 2.8$ & $x = 0.7y + 5.2$
Find mean of variables & coeff. of correlation $[\bar{x}, \bar{y}, r]$

Ans. Since LoR pass through $(\bar{x}, \bar{y})$, $(\bar{x}, \bar{y})$ satisfies given eqns.

$$\begin{aligned} \bar{y} &= 0.3\bar{x} + 2.8 \\ \bar{x} &= 0.7\bar{y} + 5.2 \end{aligned} \Rightarrow \bar{x} = 9.063, \bar{y} = 5.518$$

$$\text{Wkt } r = \sqrt{b_{xy} \times b_{yx}} = \sqrt{0.7 \times 0.3} = 0.458$$

Note • If both b_{xy} & b_{yx} are +ve $\Rightarrow r$ is +ve
• If both b_{xy} & b_{yx} are -ve $\Rightarrow r$ is -ve

Q In partially destroyed lab record, only LoR are available as $4x - 5y + 33 = 0$ & $20x - 9y = 107$. Find $\bar{x}, \bar{y}, r$

Ans. Since LoR pass through $(\bar{x}, \bar{y})$, $(\bar{x}, \bar{y})$ satisfies given eqns.

$$4\bar{x} - 5\bar{y} + 33 = 0 \Rightarrow \bar{x} = 13, \bar{y} = 17$$

$$20\bar{x} - 9\bar{y} - 107 = 0$$

LoR1 $5y = 4x + 33 \Rightarrow y = \frac{4}{5}x + \frac{33}{5} \Rightarrow b_{yx} = \frac{4}{5}$

LoR2 $9y = 20x + 107 \Rightarrow x = \frac{9}{20}y + \frac{107}{20} \Rightarrow b_{xy} = \frac{9}{20}$

$$r = \sqrt{b_{xy} \times b_{yx}} = 0.6$$

Q

x	0	1	2	3	4	5	6
y	14	13	11	9	8	5	3

Ans. $\sum x =$, $\sum y =$, $\sum xy =$, $\sum x^2 =$, $\sum y^2 =$, $\bar{x} =$, $\bar{y} =$
 $9x =$, $9y =$

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Q If $r = 0.5$ & acute angle b/w LoR is $\tan^{-1}(3/5)$. Prove $9x = 29y$ or $9y = 29x$.

Ans. $r = 0.5$, $\theta = \tan^{-1}(3/5) \Rightarrow \tan \theta = 3/5$

$$\tan \theta = \left(\frac{1 - r^2}{r} \right) \frac{9_x 9_y}{9_x^2 + 9_y^2} \Rightarrow \frac{3}{5} = \frac{3}{2} \frac{9_x 9_y}{9_x^2 + 9_y^2}$$

$$\frac{9_x 9_y}{9_x^2 + 9_y^2} = \frac{2}{5} \Rightarrow 5 9_x 9_y = 2 9_x^2 + 2 9_y^2$$

Divide by $a_x a_y \rightarrow 5 = 2 \frac{a_x}{a_y} + 2 \frac{a_y}{a_x}$

Let $\frac{a_x}{a_y} = t \Rightarrow 5 = 2t + \frac{2}{t} \Rightarrow 2t^2 - 5t + 2 = 0 \Rightarrow t = \frac{1}{2}, 2$

Case I $\frac{a_x}{a_y} = 2 \Rightarrow a_x = 2a_y$

Case II $\frac{a_x}{a_y} = \frac{1}{2} \Rightarrow 2a_x = a_y$

Multiple linear regression $\rightarrow$

Suppose y depends on 2 independent variables x_1 & x_2 , then
 $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ will be eqn. of plane, where
 $\beta_0, \beta_1, \beta_2$ are constants calculated using:

$$\sum y = \beta_0 n + \beta_1 \sum x_1 + \beta_2 \sum x_2$$

$$\sum x_1 y = \beta_0 \sum x_1 + \beta_1 \sum x_1^2 + \beta_2 \sum x_1 x_2$$

$$\sum x_2 y = \beta_0 \sum x_2 + \beta_1 \sum x_1 x_2 + \beta_2 \sum x_2^2$$

Q fit regression plane to estimate $\beta_0, \beta_1, \beta_2$ to following data of transport company on weights of 6 shipment, Distance they were moved & damage to goods incurred. Estimate damage when shipment of 3700 kg is moved 2.60 km.

Weight (x_1) (Kg)	4	3	1.6	1.2	3.4	4.8
Distance (x_2) (100km)	1.5	2.2	1	2	0.8	1.6
Damage (y) (₹)	160	112	69	90	126	186

Ans. $\sum x_1 = 18$, $\sum x_2 = 9.1$, $\sum y = 743$

$\sum x_1^2 = 63.6$, $\sum x_2^2 = 15.29$, $\sum x_1 x_2 = 27$

$\sum x_1 y = 2515.6$, $\sum x_2 y = 1133.8$

$$\begin{aligned}
 743 &= 6\beta_0 + 18\beta_1 + 9.1\beta_2 \\
 2515.6 &= 18\beta_0 + 63.6\beta_1 + 27\beta_2 \\
 1133.8 &= 9.1\beta_0 + 27\beta_1 + 15.29\beta_2
 \end{aligned}
 \Rightarrow \beta_0 = 16.987, \beta_1 = 30.189, \beta_2 = 10.732$$

$$\therefore y = 16.987 + 30.189x_1 + 10.732x_2$$

$$\text{Now, } x_1 = 3.7, x_2 = 2.6 \Rightarrow y = 156.589$$

Q Find regression plane of y on x_1 & x_2 , given $\sum \bar{x}_{1i}^2 = 38.4$, $\sum \bar{x}_{2i}^2 = 3.4$, $\sum \bar{x}_{1i}\bar{y}_i = 29.76$, $\sum \bar{x}_{2i}\bar{y}_i = 8.94$, $\sum \bar{x}_{1i}\bar{x}_{2i} = 9.6$, where $\bar{y}_i = y_i - \bar{y}$, $\bar{x}_{1i} = x_{1i} - \bar{x}_1$, $\bar{x}_{2i} = x_{2i} - \bar{x}_2$ are deviations of data from their means. $\bar{y} = 55.2$, $\bar{x}_1 = 20.1$, $\bar{x}_2 = 6.4$

Ans.

Consider regression plane $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ --- (i)
 Since $(\bar{x}_1, \bar{x}_2, \bar{y})$ will lie on regression plane, we have:
 $\bar{y} = \beta_0 + \beta_1 \bar{x}_1 + \beta_2 \bar{x}_2$ --- (ii)

$$(i) - (ii) \quad y - \bar{y} = \beta_1 (x_1 - \bar{x}_1) + \beta_2 (x_2 - \bar{x}_2) \Rightarrow \bar{y}_i = \beta_1 \bar{x}_{1i} + \beta_2 \bar{x}_{2i}$$

$$\sum \bar{x}_{1i} \bar{y}_i = \beta_1 \sum \bar{x}_{1i}^2 + \beta_2 \sum \bar{x}_{1i} \bar{x}_{2i}$$

$$\sum \bar{x}_{2i} \bar{y}_i = \beta_1 \sum \bar{x}_{1i} \bar{x}_{2i} + \beta_2 \sum \bar{x}_{2i}^2$$

$$29.76 = 38.4\beta_1 + 9.6\beta_2 \Rightarrow \beta_1 = 0.4, \beta_2 = 1.5$$

$$8.94 = 9.6\beta_1 + 3.4\beta_2$$

Substitute in (ii) $\rightarrow$

$$55.2 = \beta_0 + 0.4 \times 20.1 + 1.5 \times 6.4 \Rightarrow \beta_0 = 37.56$$

$$\therefore y = 37.56 + 0.4x_1 + 1.5x_2$$

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Random variable $\rightarrow$

- Real variable X associated with outcome of random experiment
- Also called stochastic variable or just variable

eg. Tossing pair of dice

 X : ~~less~~ Sum of 2 no.s which is even

$$X = \{11, 22, 33, 44, 55, 66, 13, 15, 24, 26, 31, 35, 42, 46, 51, 53, 62, 64\} = \{2, 4, 6, 8, 10, 12\}$$

- If random variable takes finite value $\Rightarrow$ Discrete
- If random variable takes continuous value $\Rightarrow$ Continuous infinite

Discrete Probability Distribution $\rightarrow$

- Let X be random variable
- Prob. that X takes value x_i
 $p(x) = p(x_i) = p_i$, $\forall i = 1, 2, 3, \dots, n$, [prob. mass function]
 where

$$0 \leq p(x_i) \leq 1$$

$$\sum_{i=1}^n p(x_i) = 1$$

- Set of (x_i, p_i) constitutes discrete prob. distribution

eg. Tossing pair of dice

 X : Sum of no.s

$$X = \{2, 3, \dots, 12\}$$

$$p(X) =$$

$X =$	2	3	4	5	6	7	8	9	10	11	12
$p(X) =$	$1/36$	$2/36$	$3/36$	$4/36$	$5/36$	$6/36$	$5/36$	$4/36$	$3/36$	$2/36$	$1/36$

Note: Above table $\Rightarrow$ DPD

$$P(X=8) = 5/36$$

$$P(X \leq 3) = \frac{1}{36} + \frac{2}{36} = 3/36 = 1/12$$

CDF Cumulative Distribution Function $\rightarrow$

$F(x)$

$$F(x_k) = p(x \leq x_k) = \sum_{i=1}^k p(x_i)$$

eg $F(8) = p(x \leq 8) = p(8) + p(7) + p(6) + p(5) + p(4) + p(3) + p(2) + p(1)$

Continuous Prob. Distribution $\rightarrow$

If X takes value in range $[a, b]$, then prob. that X takes value constitutes this CPD.

Distribution function $\rightarrow f_b(x) \rightarrow$ prob. density function

$$* f_b(x) \geq 0$$

$$* \int_{-\infty}^{\infty} f_b(x) dx = 1$$

$$* p(a \leq x \leq b) = \int_a^b f_b(x) dx$$

Note $p(a < x < b) = p(a \leq x \leq b) \Rightarrow$ Conclusion of endpoints doesn't affect area under curve

However, this is not case in discrete prob. dist.

2/5 Cumulative prob. Dist. →

In continuous prob. dist., cumulative prob. dist. function:

$$F(c) = \int_{-\infty}^c f(x) dx$$

$$p(a \leq x \leq b) = \int_{-\infty}^b f(x) dx - \int_{-\infty}^a f(x) dx = F(b) - F(a)$$

Mean, variance & standard deviation of prob. dist. →

• Mean → Also called expectation $E(X)$

$$\text{Mean } (\mu \text{ or } E(X)) \rightarrow \sum x_i p(x_i)$$

[Discrete]

$$\text{Variance } (\sigma^2) \rightarrow \sum (x_i - \mu)^2 p(x_i)$$

$$\int_{-\infty}^{\infty} x f(x) dx$$

[Continuous]

$$\int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

Q Prove that $\sigma^2 = E(X^2) - (E(X))^2$

$$\text{Ans. } \sigma^2 = \sum (x - \mu)^2 p(x) = \sum (x^2 + \mu^2 - 2\mu x) p(x)$$

$$= \sum x^2 p(x) + \mu^2 \sum p(x) - 2\mu \sum x p(x)$$

$$= E(X^2) + \mu^2 - 2\mu^2 = E(X^2) - \mu^2$$

$$= E(X^2) - (E(X))^2$$

Q Prob. dist. of X is given by:

X	0	1	2	3	4	5
$p(X)$	k	$5k$	$10k$	$10k$	$5k$	k

Find k , $E(X)$, σ^2 .

Ans. $\sum p(x_i) = 1 \Rightarrow 32k = 1 \Rightarrow k = 0.03125 = 1/32$

$$E(X) = \sum x_i p(x_i) = 0 \times k + 1 \times 5k + 2 \times 10k + 3 \times 10k + 4 \times 5k + 5 \times k$$

$$= 80k = 80 \times \frac{1}{32} = 2.5$$

$$\sigma^2 = E(X^2) - (E(X))^2$$

$$E(X^2) = \sum x_i^2 p(x_i)$$

$$= 0^2 \times k + 1^2 \times 5k + 2^2 \times 10k + 3^2 \times 10k + 4^2 \times 5k + 5^2 \times k$$

$$= 240k = 240 \times \frac{1}{32} = 7.5$$

$$\sigma^2 = 7.5 - (2.5)^2 = 1.25$$

Q

X	1	2	3	4	5	6	7
p(X)	k	2k	2k	3k	k ²	2k ²	7k ² + k

Find k , $p(x \geq 6)$, $p(1 \leq x \leq 5)$, $p(x > 6)$, $E(X)$, σ^2 .

Ans.

$$\sum p(x_i) = 1 \Rightarrow 10k^2 + 9k = 1 \Rightarrow k = 0.1, -1 = 0.1$$

$$p(x \geq 6) = k + 2k + 2k + 3k + k^2 + 2k^2 = 3k^2 + 8k = 0.83$$

$$p(1 \leq x \leq 5) = k + 2k + 2k + 3k + k^2 = k^2 + 8k = 0.81$$

$$p(x \geq 6) = 2k^2 + 7k^2 + k = 9k^2 + k = 0.19$$

$$p(x > 6) = 7k^2 + k = 0.17$$

$$E(X^2) = \sum x_i^2 p(x_i)$$

$$= 1^2 \times k + 2^2 \times 2k + 3^2 \times 2k + 4^2 \times 3k + 5^2 \times k^2 + 6^2 \times 2k^2 + 7^2 \times (7k^2 + k)$$

$$= 440k^2 + 124k = 16.8$$

$$E(X) = \sum x_i p(x_i) = 6k^2 + 30k = 3.6$$

$$\sigma^2 = E(X^2) - (E(X))^2 = 16.8 - (3.6)^2 = 3.56$$

Q Prob. dens. fn. of random variable

$$f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

Find ~~later~~ cumulative prob. dist. fn., $p(x \geq 1.5)$

Ans.

$$* F(x), \text{ when } x < 0 = \int_{-\infty}^x f(x) dx = 0$$

$$* F(x), \text{ when } 0 < x < 1 = \int_{-\infty}^x f(x) dx = \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx$$

$$= \int_0^x x dx = \left[\frac{x^2}{2} \right]_0^x = \frac{x^2}{2}$$

$$* F(x), \text{ when } 1 < x < 2 = \int_{-\infty}^x f(x) dx$$

$$= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^x f(x) dx$$

$$= \int_0^1 x dx + \int_1^x (2-x) dx$$

$$= \left[\frac{x^2}{2} \right]_0^1 + \left[2x - \frac{x^2}{2} \right]_1^x = 2x - \frac{x^2}{2} - 1$$

$$F(x) = \begin{cases} 0, & x < 0 \\ x^2/2, & 0 \leq x \leq 1 \\ 2x - x^2/2 - 1, & 1 \leq x \leq 2 \\ 1, & x > 2 \end{cases}$$

$$* F(x), \text{ when } x > 2 = \int_{-\infty}^x f(x) dx$$

$$= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^x f(x) dx$$

$$p(x \geq 1.5) = 1 - F(1.5) = 1 - (2(1.5) - \frac{(1.5)^2}{2} - 1) = -4.875 + 0.125$$

Q Prob. dens. fn. of random variable $f(x) = \begin{cases} Kx^2, & -3 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$

Find K , $p(1 \leq x \leq 2)$, $p(|x| > 1)$, $p(2x+3 > 5)$, $E(X)$, σ^2 .

Ans.

$$\text{Wkt } \int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_{-\infty}^{-3} 0 dx + \int_{-3}^3 Kx^2 dx + \int_3^{\infty} 0 dx = 1$$

$$K \left[\frac{x^3}{3} \right]_{-3}^3 = 1 \Rightarrow K = 1/18$$

$$p(1 \leq x \leq 2) = \int_1^2 f(x) dx = \int_1^2 Kx^2 dx = K \left[\frac{x^3}{3} \right]_1^2 = \frac{7}{54}$$

$$p(|x| > 1) = p(x < -1) + p(x > 1)$$

Note $|x| < 1 \Rightarrow -1 < x < 1$

$$|x| > 1 \Rightarrow (-\infty < x < -1) \& (1 < x < \infty)$$

$$p(|x| > 1) = p(x < -1) + p(x > 1) = \int_{-\infty}^{-1} f(x) dx + \int_1^{\infty} f(x) dx$$

or

$$p(|x| > 1) = 1 - p(|x| < 1) = 1 - p(-1 < x < 1)$$

$$= 1 - \int_{-1}^1 f(x) dx = 1 - \int_{-1}^1 Kx^2 dx$$

$$= 1 - K \left[\frac{x^3}{3} \right]_{-1}^1 = 1 - \frac{1}{27} = \frac{26}{27} = 0.9629$$

$$p(2x+3 > 5) = p(2x > 2) = p(x > 1) = \int_1^{\infty} f(x) dx$$

$$= \int_1^3 Kx^2 dx + \int_3^{\infty} 0 dx = K \left[\frac{x^3}{3} \right]_1^3 = \frac{13}{27} = 0.481$$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_{-3}^3 x Kx^2 dx = 0 \quad [\text{Even fn.}]$$

$$E(X^2) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx = \int_{-\infty}^{\infty} x^2 \cdot f(x) dx$$

$$= \int_{-3}^3 x^2 \cdot kx^2 dx = k \left[\frac{x^5}{5} \right]_{-3}^3 = \frac{27}{18} = \frac{3}{2}$$

$$\sigma^2 = E(X^2) - (E(X))^2$$

$$= (1.5) - (0)^2 = 1.5 \quad (0 \leq x \leq 1)$$

3/5

Q Find $E(X)$, $E(X^2)$ & σ^2 for prob. dist $p(x)$ defined by [where k is appropriate constant.]

x	1	2	3	...	n
$p(x)$	k	$2k$	$3k$	...	nk

Ans.

$$\text{Wkt } \sum p(x) = 1 \Rightarrow k(1+2+3+\dots+n) = 1 \Rightarrow k \left(\frac{n(n+1)}{2} \right) = 1 \Rightarrow k = \frac{2}{n(n+1)}$$

$$E(X) = \sum X p(X) = 1 \times k + 2 \times 2k + 3 \times 3k + \dots + n \times nk$$

$$= k(1^2 + 2^2 + 3^2 + \dots + n^2)$$

$$= k \frac{n(n+1)(2n+1)}{6} = \frac{2}{n(n+1)} \times \frac{n(n+1)(2n+1)}{6}$$

$$= \frac{2n+1}{3}$$

$$E(X^2) = \sum X^2 p(X)$$

$$= 1^2 \times k + 2^2 \times 2k + 3^2 \times 3k + \dots + n^2 \times nk$$

$$= k(1^3 + 2^3 + 3^3 + \dots + n^3)$$

$$= k \left(\frac{n(n+1)}{2} \right)^2 = \frac{n(n+1)}{2}$$

$$\sigma^2 = E(X^2) - (E(X))^2 = \frac{n(n+1)}{2} - \left(\frac{2n+1}{3} \right)^2$$

prob. Binomial Distribution $\rightarrow$

- * Experiment is repeated 'n' no. of times called 'n' trials
- * Outcome of each trial is classified into 2 mutually exclusive events $\rightarrow$ failure & success
- * prob. of success (p) remains constant for all trials
- * Outcomes are different

prob. mass fn. $\rightarrow p(X=r) = {}^n C_r p^r q^{n-r}$

(n) no. of trials

p prob. (success)

q prob. (failure)

r no. of successes

$$\begin{aligned} \text{Consider } \sum p(X_i) &= p(X=0) + p(X=1) + p(X=2) + \dots + p(X=n) \\ &= {}^n C_0 p^0 q^{n-0} + {}^n C_1 p^1 q^{n-1} + {}^n C_2 p^2 q^{n-2} + \dots + {}^n C_n p^n q^{n-n} \\ &= (p+q)^n = 1^n = 1 \end{aligned}$$

Mean & Variance of Binomial dist. $\rightarrow$

$$\begin{aligned} \text{Mean} &= E(X) = \sum X p(X) = \sum x {}^n C_x p^x q^{n-x} \quad [p(X=x)] \\ &= \sum x \frac{n!}{(n-x)!x!} p^x q^{n-x} \\ &= \sum \frac{n!}{(n-x)!(x-1)!} p^x q^{n-x} = \sum \frac{n!}{(n-x)!(x-1)!} p^{x-1} p q^{n-x} \end{aligned}$$

$$\text{Put } x-1 = h \Rightarrow x = h+1$$

$$p \sum \frac{n!}{(n-h-1)!(h)!} p^h q^{n-h-1} = p \sum \frac{n(n-1)!}{((n-1)-h)!h!} p^h q^{(n-1)-h}$$

$$= np \leq \frac{(n-1)!}{((n-1)-h)!h!} p^h q^{(n-1)-h}$$

$$= np \leq {}^m C_h p^h q^{m-h}, \text{ where } m = n-1$$

$$* E(X) = np \leq \sum p(X=h) = np$$

$$E(X^2) = \sum x^2 p(x)$$

$$\text{Let } x^2 = x(x-1) + x$$

$$E(X^2) = \sum [x(x-1) + x] p(x) = \sum x(x-1) p(x) + \sum x p(x)$$

$$= \sum x(x-1) p(x) + np \quad \text{--- (ii)}$$

$$\text{Consider } \sum x(x-1) p(x) = \sum x(x-1) {}^n C_x p^x q^{n-x}$$

$$= \sum x(x-1) \frac{n!}{(n-x)!x!} p^x q^{n-x} = \sum \frac{n!}{(n-x)!(x-2)!} p^x q^{n-x}$$

$$\left[\frac{x(x-1)}{x!} = \frac{x(x-1)}{x(x-1)(x-2)!} \right]$$

$$= \sum \frac{n!}{(n-x)!(x-2)!} p^{x-2} p^2 q^{n-x} = p^2 \sum \frac{n!}{(n-x)!(x-2)!} p^{x-2} q^{n-x}$$

$$\text{Put } x-2 = h \Rightarrow x = 2+h$$

$$= p^2 \sum \frac{n(n-1)(n-2)!}{((n-2)-h)!h!} p^h q^{(n-2)-h}$$

$$= p^2 n(n-1) \sum \frac{(n-2)!}{((n-2)-h)!h!} p^h q^{(n-2)-h}$$

$$= p^2 n(n-1) \sum p(X=h)$$

$$\sum x(x-1) p(x) = p^2 n(n-1)$$

$$\text{iii) } E(X^2) = p^2 n(n-1) + np$$

$$* \quad \sigma^2 = p^2 n(n-1) + np^2$$

$$= np(1+p) = npq$$

6/5

$$Q \quad \mu \rightarrow \frac{15}{4}, \sigma^2 \rightarrow \frac{15}{6}, n = ?$$

$$\text{Ans. } \mu = E(X) = np = \frac{15}{4}$$

$$\sigma^2 = npq = \frac{15}{6} \Rightarrow q = \frac{2}{3} \Rightarrow p = \frac{1}{3} \Rightarrow n = \frac{15}{4} = 11.25$$

Q Mean & variance of binomial variable X with parameter (n, p) are 16 & 8. Find dist. of X .

$$\text{Ans. } \mu = E(X) = np = 16$$

$$\sigma^2 = npq = 8 \Rightarrow q = \frac{1}{2} \Rightarrow p = \frac{1}{2} \Rightarrow n = 32$$

$$P(X=n) = {}^nC_n p^n q^{n-n} = {}^{32}C_n \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^{32-n} = {}^{32}C_n \left(\frac{1}{2}\right)^{32}$$

Q prob. that man aged 60 will live upto 70 is 0.65. Find prob. that out of 10 men now aged 60, atleast 7 will live upto 70.

Ans. Success : Man aged 60 will live upto 70 $\Rightarrow p(\text{Success}) = 0.65$

$$n = 10$$

$$q = 1 - p = 0.35$$

$$\begin{aligned} P(X \geq 7) &= P(X=7) + P(X=8) + P(X=9) + P(X=10) \\ &= {}^{10}C_7 (0.65)^7 (0.35)^3 + {}^{10}C_8 (0.65)^8 (0.35)^2 + {}^{10}C_9 (0.65)^9 (0.35)^1 \\ &\quad + {}^{10}C_{10} (0.65)^{10} (0.35)^0 \\ &= 0.252 + 0.175 + 0.072 + 0.013 \\ &= 0.512 \end{aligned}$$

Q During war, 1 ship out of 9 sunk on avg. in making certain voyage. Find prob. that 3 out of 6 ships would arrive safely.

Ans. Success: Ship arrives safely. $\Rightarrow p(\text{success}) = 8/9 \Rightarrow q = 1/9$
 $n = 6$

$$P(X=3) = {}^6C_3 (8/9)^3 (1/9)^3 = 0.019$$

Q Find prob. of getting 9 exactly twice in 3 throws with pair of fair dice.

Ans. Success: Getting sum of 9 in pair of dice $\Rightarrow p(\text{success}) = 4/36 = 1/9$

$$n = 3$$

$$q = \frac{8}{9}$$

$$P(X=2) = {}^3C_2 \left(\frac{1}{9}\right)^2 \left(\frac{8}{9}\right)^1 = 0.0329$$

Q Out of 800 families, with 5 children each, how many are expected to have (i) 3 boys (ii) 5 girls (iii) either 2 or 3 boys? Assume equal prob. for boys & girls

Ans. Success: Child is boy $\Rightarrow p(\text{success}) = 0.5 \Rightarrow q = 0.5$
 $n = 5$

$$\text{i) } P(X=3) = {}^5C_3 (0.5)^3 (0.5)^2 = 0.3125 \times 800 = 250$$

$$\text{ii) } P(X=0) = {}^5C_0 (0.5)^0 (0.5)^5 = 0.03125 \times 800 = 25$$

$$\text{iii) } P(X=2) + P(X=3) = {}^5C_2 (0.5)^2 (0.5)^3 + {}^5C_3 (0.5)^3 (0.5)^2 = 0.3125 + 0.3125 = 0.625 \times 800 = 500$$

9/5

Q An airline knows that 5% of ppl making reservations will not turn up. Their policy is to sell 52 tickets for 50-seat flight. Find prob. that there will be seat for every passenger who turns up.

Ans. Success : passenger who makes reservation will ^{not} turn up
 $p(\text{failure}) = 5\% = 0.05 = p \Rightarrow q = 0.95 = q$
 $n = 52$

Since flight can hold 50 passengers, atleast 2 people shouldn't turn up for everyone to get a seat.

$$P(X \geq 2) = 1 - P(X < 2) = 1 - P(X=0) - P(X=1)$$

$$= 1 - {}^{52}C_0 (0.05)^0 (0.95)^{52} - {}^{52}C_1 (0.05)^1 (0.95)^{51}$$

$$= 0.740$$

Poisson distribution $\rightarrow$

- Limiting case of binomial dist.
- Arises when no. of trials increases indefinitely & product $np = E(X)$ remains constant

Consider prob. mass. fn of binomial dist. $\rightarrow$

$$P(X=h) = {}^nC_h p^h q^{n-h} = \frac{n!}{(n-h)!h!} \left(\frac{\mu}{n}\right)^h \left(1 - \frac{\mu}{n}\right)^{n-h} \quad [\mu = np]$$

Take limit as $n \rightarrow \infty \Rightarrow \lim_{n \rightarrow \infty} \frac{n(n-1)(n-2)\dots((n-h)+1)(n-h)!}{(n-h)!h!} \left(\frac{\mu}{n}\right)^h \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-h}$

$$\lim_{n \rightarrow \infty} \frac{n(n-1)(n-2)\dots((n-h)+1)(n-h)!}{(n-h)!h!} \left(\frac{\mu}{n}\right)^h \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-h}$$

$$= \lim_{n \rightarrow \infty} \frac{n(n-1)(n-2)\dots(n-h+1)}{n^h} \frac{\mu^h}{h!} \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-h}$$

$$= \lim_{n \rightarrow \infty} \frac{1 \cdot \left(1 - \frac{1}{n}\right) \cdot \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{h-1}{n}\right)}{1} \frac{\mu^h}{h!} \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-h}$$

$$P(X=h) = \frac{\mu^h}{h!} e^{-\mu}, \text{ where } \mu \rightarrow \text{parameter of pois. dist.}$$

Note: $\mu > 0$ [$h = 0, 1, 2, \dots \rightarrow$ no. of successes]

- In statistics, if no. of samples $\geq 30 \Rightarrow$ large sample & we consider pois. dist. instead of binomial dist.

Mean & Variance of poisson distribution $\rightarrow$

$$E(X) = \sum x p(x) \\ = \sum x \cdot \frac{\mu^x}{x!} e^{-\mu}$$

$$[p(X=x) = p(x)]$$

$$= \sum x \cdot \frac{\mu^x}{x(x-1)!} e^{-\mu} = \mu e^{-\mu} \sum \frac{\mu^{x-1}}{(x-1)!}$$

$$= \mu e^{-\mu} \sum \frac{\mu^{x-1}}{(x-1)!} = \mu e^{-\mu} \left[1 + \mu + \frac{\mu^2}{2!} + \frac{\mu^3}{3!} + \dots \right]$$

$$= \mu e^{-\mu} \cdot e^{\mu} = \mu$$

$$\therefore E(X) = \mu$$

$$V^2 = E(X^2) - (E(X))^2 = E(X^2) - \mu^2$$

$$\text{Consider } E(X^2) = \sum x^2 p(x)$$

$$\begin{aligned}
 E(X) &= \sum (x(x-1) + x) p(x) = \sum x(x-1) p(x) + \sum x p(x) \\
 &= \sum x(x-1) p(x) + \mu \\
 &= \sum x(x-1) \frac{\mu^x e^{-\mu}}{x!} + \mu = \sum x(x-1) \frac{\mu^{x-2} \mu^2 e^{-\mu}}{x(x-1)(x-2)!} + \mu \\
 &= \mu^2 e^{-\mu} \sum \frac{\mu^{x-2}}{(x-2)!} + \mu = \mu^2 e^{-\mu} e^{\mu} + \mu
 \end{aligned}$$

$$E(X^2) = \mu^2 + \mu$$

$$\sigma^2 = (\mu^2 + \mu) - (\mu)^2 = \mu$$

Note: Can also use m in place of μ

Q If random variable X has pois. dist. such that $p(X=1) = p(X=2)$
Find Mean & σ^2 .

Ans.

$$P(X=n) = \frac{m^n e^{-m}}{n!}$$

$$P(X=1) = P(X=2) \Rightarrow \frac{m^1 e^{-m}}{1!} = \frac{m^2 e^{-m}}{2!} \Rightarrow m = 2$$

$$E(X) = \sigma^2 = m$$

Q Prob. of pois. variate taking value 3 & 4 are equal.
Find prob. of taking value 0 & 1.

Ans.

$$P(X=n) = \frac{m^n e^{-m}}{n!}$$

$$P(X=3) = P(X=4) \Rightarrow \frac{m^3 e^{-m}}{3!} = \frac{m^4 e^{-m}}{4!} \Rightarrow m = 4$$

$$P(X=0) = \frac{m^0 e^{-m}}{0!} = e^{-4}$$

$$P(X=1) = \frac{m^1 e^{-m}}{1!} = 4e^{-4}$$

Q. 2% of fuses made by firm are found to be defective. Find prob. that box containing 200 fuses contains (i) no defective (ii) 3 or more defective fuses.

Ans. $p = 0.02$, $n = 200 \Rightarrow \mu = np = 4 = m$

$$P(X=r) = \frac{m^r e^{-m}}{r!}$$

$$i) P(X=0) = \frac{m^0 e^{-m}}{0!} = \frac{4^0 e^{-4}}{0!} = e^{-4} = 0.018$$

$$ii) P(X \geq 3) = 1 - P(X=3) - P(X=2) - P(X=1) - P(X=0)$$

$$= 1 - \frac{4^3 e^{-4}}{3!} - \frac{4^2 e^{-4}}{2!} - \frac{4^1 e^{-4}}{1!} - \frac{4^0 e^{-4}}{0!}$$

$$= 1 - \frac{32 e^{-4}}{3} - \frac{8 e^{-4}}{2} - 4 e^{-4} - e^{-4}$$

$$= 1 - \frac{32 e^{-4}}{3} - 4 e^{-4} - e^{-4} - e^{-4}$$

$$= 1 - \frac{32 e^{-4}}{3} - 6 e^{-4}$$

$$= 0.7618$$

Q Bag contains 1 red & 7 white balls. Ball is drawn & colour is observed. Ball is put back & contents are mixed. Using

(i) Binomial dist. (ii) pois. approx. to binomial dist., find prob that in 8 such drawings, red is picked exactly 3 times.

Ans. $n = 8$, $r = 3$, $p = 1/8$, $q = 7/8$

$$i) P(X=r) = {}^n C_r p^r q^{n-r} = {}^8 C_3 p^3 q^5 = \frac{8 \times 7 \times 6!}{3! \times 5!} \left(\frac{1}{8}\right)^3 \left(\frac{7}{8}\right)^5 = 0.056$$

$$ii) P(X=r) = \frac{\mu^r e^{-\mu}}{r!} = \frac{1^3 e^{-1}}{3!}$$

$$= 0.061$$

14/5

Q Communication channel receives independent pulses @ 12p/ms. Prob. of transmission error is 0.001 for each ms. Find prob. of (i) no error (ii) ~~at least 1~~ (iii) ~~at least 1~~ (iii) 2 (iv) ~~at least 2~~

Ans. $n = 12$

Success : error $\Rightarrow p(\text{success}) = 0.001$
 $m = np = 0.012$

$$P(X=n) = \frac{m^n e^{-m}}{n!}$$

$$i) P(X=0) = \frac{(0.012)^0 e^{-0.012}}{0!} = 0.988$$

$$ii) P(X \geq 1) = 1 - P(X=1) = 1 - \frac{(0.012)^1 e^{-0.012}}{1!} = 0.988$$

$$iii) P(X=2) = \frac{(0.012)^2 e^{-0.012}}{2!} = 7.114 \times 10^{-5}$$

$$iv) P(X \leq 2) = P(X=0) + P(X=1) + P(X=2) = 0.988 + 0.0118 + 7.114 \times 10^{-5} = 0.9998 \approx 1$$

Q No. of monthly breakdowns of computer is random variable having pois. dist. with mean 1.8. Find prob. that computer will function for a month (i) without breakdown (ii) only 1 (iii) at least 1

Ans. $m = 1.8$

Success : breakdown

$$i) P(X=0) = \frac{(1.8)^0 e^{-1.8}}{0!} = 0.165$$

$$ii) P(X=1) = \frac{(1.8)^1 e^{-1.8}}{1!} = 0.297$$

$$iii) P(X \geq 1) = 1 - P(X=0) = 1 - 0.165 = 0.835$$

Q. It is pois. dist. for: x 0 1 2 3 4

f 122 60 15 2 1

Ans. $m = \frac{\sum x_i f_i}{\sum f_i}$

$\sum f_i = 122 + 60 + 15 + 2 + 1 = 200$

$$m = \frac{100}{200} = 0.5$$

x 0 1 2 3 4

~~0.0758~~

x 0 1 2 3 4

P 0.6065 0.303 0.0758 0.0126 0.0015

$f_i P_i$ 121.3 60.648 15.16 2.52 0.315

$$E(X) = \frac{121.3 + 60.648 + 15.16 + 2.52 + 0.315}{200} = 0.5$$

$$E(X^2) = \frac{121.3 + 60.648 + 15.16 + 2.52 + 0.315}{200} = 0.5$$

$$E(X^2) = \frac{121.3 + 60.648 + 15.16 + 2.52 + 0.315}{200} = 0.5$$

$$1 = 800P_0 =$$

Let us assume that the number of successes in n trials is X .

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Let us assume that the number of successes in n trials is X .

$E(X) = nP$

$E(X^2) = nP$

$$E(X) = nP = 10 \times 0.5 = 5$$

$$E(X^2) = nP = 10 \times 0.5 = 5$$

$$E(X^2) = nP = 10 \times 0.5 = 5$$